

Grain initialization for the k_{eff} upscaling runs

enceladus_DSM

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The large-domain runs behind `studies/keff_sintering/` start from a *constructed* granular microstructure, not from an equilibrated field and not from a mechanical settling simulation. Initialization happens in two clearly separated stages:

1. **Offline (Python).** A sharp geometry — a list of non-overlapping circles (x_k, y_k, R_k) — is generated, *graded*, and committed to the repository as `inputs/packings/.../grains.dat`. No phase field, no ϵ , no mesh is involved; this stage is temperature-independent and is reused at every run temperature.
2. **Online (solver).** At $t = 0$ the solver reads that list and paints a diffuse phase field over it, together with the temperature and vapour-density fields. This stage is where ϵ and the mesh enter.

Keeping the two apart is the design decision the rest of this note elaborates: the sharp geometry is fixed and auditable, and ϵ only sets the thickness of the diffuse skin painted around it.

1 What the packings are meant to be

Every k_{eff} packing targets the same microstructure and differs only in box size and random seed:

porosity ϕ	0.325	gated to ± 0.01
mean grain radius R_{ave}	50 μm	log-normal, $\sigma_{\ln} = 0.5$
radius clip	$[0, 2R_{ave}]$	-radius-clip-frac 1.0
periodicity	both axes (xy)	the only mode with no boundary layer
dimension	2	

Two box sizes are in use, both set by the RVE ratio L/R_{ave} rather than by a length chosen directly:

pilot:	$L/R_{ave} = 40$	$L = 2.0 \text{ mm}$	<code>inputs/packings/pilot_LR40/</code>
REV:	$L/R_{ave} = 64$	$L = 3.2 \text{ mm}$	<code>inputs/packings/rev_LR64/</code>

with four seeds each, so seed scatter is measured rather than assumed. Periodicity in *both* axes is required: under `-periodic 0` the zero-flux wall imposes a 90° contact angle on ϕ , so a grain clipped by a wall carries the wrong surface curvature and coarsens as though it were smaller than it is.

2 Stage 1 (offline): building the sharp packing

`preprocess/generate_packing_gravity.py`. Five steps, in order.

1.1 Geometric drop-and-roll deposition. Grains are placed one at a time into a strip of width L . A radius is drawn from the log-normal, an x is drawn uniformly, and the grain is lowered until it first touches the floor or a settled grain; it then *rolls downhill* around that contact until it finds a second support, exhausts a rolling budget, reaches the floor, or rolls past the support’s equator (in which case it falls off and is re-dropped, up to four hops).

There is no DEM, no contact law and no time integration — this must not be reported as a settling simulation. What it buys is the one structural property that matters here: every deposited grain rests on something below it, so the solid matrix is connected *by construction* rather than by luck, and no placement that overlaps a settled grain is accepted.

1.2 Porosity by bisecting a physical knob. The rolling budget `-roll-tol` is the porosity control: 0 means lock on first touch (ballistic, $\phi \approx 0.45\text{--}0.50$) and $\pi/2$ means roll to a stable two-point seat every time (dense, $\phi \approx 0.16\text{--}0.20$). Porosity falls monotonically with the budget, so it is bisected *per seed* onto the target. Radii are never rescaled to hit a porosity: rescaling would break contact tangency, which is the whole point (see *Why additive* in §3). The realized budgets for the eight packings in use run 0.94–1.26 rad.

Deposition always wraps in x regardless of the requested output periodicity: the bed being modelled is wide, and a strip with free sides opens spurious voids at the edges and drives the interior too dense.

1.3 The domain is a window cut from a larger bed. Deposition runs in a strip taller than the domain by $3R_{ave}$ above and below, and the output window is cropped from its interior. Both discarded layers are artifacts: the floor contact layer at the bottom and the loose, half-filled free surface at the top. Deposition stops on the *minimum* of the bed surface profile, not the highest grain, so the crop never reaches up into that loose layer.

1.4 Seam healing, then filling — in that order. Identifying $y = 0$ with $y = L_y$ leaves grains near the seam overlapping, so the seam is relaxed apart first (`packing_lib.relax` under the minimum image, with x wrapped). Maximum grain displacement is recorded as `seam_max_shift_m`; it is 28–81 μm , i.e. confined to grains at the seam.

The deposited skeleton is deliberately bisected to land 0.03 *above* the target porosity, and the remainder is removed by inserting grains into the largest voids, biggest hole first. Attacking the largest inscribed empty circle serves two goals with one pass: porosity falls to target and pore heterogeneity falls with it. Each insert is capped by the largest circle that fits *and* by the remaining area deficit, and the last one is shrunk to land exactly on target rather than overshoot.

Filler grains are not required to rest on anything — some float. That is intentional: a 2D section through a 3D packing shows unsupported grains all the time, and a grain connected out of plane is mechanically real even where the slice cannot show it. The matrix is built connected *first* so the floaters garnish a percolating skeleton instead of constituting it, and `metadata.json` records the split (`n_matrix_grains` vs `n_filler_grains`: 4–6% filler across the eight packings).

1.5 Acceptance is a gate, not a ranking. For homogenization the domain has to be a usable REV, so a packing that is well connected on one side and sparse on the other is worse than useless: it yields a number that looks fine and means nothing. The generator therefore *refuses* to emit a packing and re-seeds (up to 24 times) unless all of

- the largest inscribed pore circle is below its gate,

- the local density CV on a coarse grid is below 0.3475,
- the half-domain solid-fraction difference is below 22% in either axis,
- the solid percolates in *both* x and y ,
- achieved porosity is within 0.01 of target

hold. Three of the four pilot packings needed 2–3 attempts. Packings are never *ranked* on pore connectivity: maximising the fraction of void in one cluster selects for clumped grains beside one large hole, which is the opposite of representative.

The void gate is applied per domain size, not in mean radii: the largest of N voids grows with N , so a flat mean-radii threshold calibrated on the $L/R_{ave} = 40$ box silently rejects the larger and relatively *more* uniform $L/R_{ave} = 64$ box.¹

1.6 What is written out. `grains.dat` (an $L_x L_y$ header then $x y r$ rows, all in metres), `metadata.json` (every graded metric, the gates, the seed and attempt, and a SHA-256 of `grains.dat`) and `preview.png`.

One detail in `grains.dat` matters more than its size suggests: **periodic edge images are written explicitly**. Any grain reaching past a periodic face is emitted a second time at its image, so centres span $[-R, L + R]$ and a grain need only be *partly* inside the domain. Dropping the edge-straddling grains would sever exactly the conduction paths that leave the domain, and connectivity across the boundary is what k_{eff} is most sensitive to — it would bias k_{eff} low. The images also make the microstructure genuinely periodic across those faces, which is what the corrector problem’s periodic boundary conditions assume. The pilot files carry 25–30 images on top of 308–336 distinct grains.

3 Stage 2 (online): the fields at $t = 0$

`src/initial_conditions.c`, reached via `-ic_type multi_grains_file` and `-grains_file <path>`. The grain list is parsed on rank 0 and broadcast, so every rank sees byte-identical geometry; a ragged or zero-radius file is an error rather than a silent misinterpretation.

2.1 The ice phase field: additive, clamped tanh profiles. ϕ is the clamped *sum* of the grains’ individual diffuse fields (`-ic_grain_union 0`, the Molaro convention):

$$\phi(\mathbf{x}) = \text{clamp}\left(\sum_k \left[\frac{1}{2} - \frac{1}{2} \tanh\left(\frac{r_k - R_k}{2\epsilon}\right)\right], 0, 1\right), \quad r_k = \|\mathbf{x} - \mathbf{c}_k\|. \quad (1)$$

The tanh coefficient is $1/(2\epsilon)$, i.e. the model’s own equilibrium logistic profile $\phi = 1/(1 + e^{-x/\epsilon})$. Initializing at the equilibrium width removes the early width-relaxation transient that a steeper initial profile would spend its first ~ 60 steps undoing. Note that ϵ is a *decay length*: the visible band is $\approx 6\epsilon$ (5%–95%) to $\approx 9.2\epsilon$ (1%–99%), not ϵ .

¹Visible in the metadata: REV seed 1 records `gate_max_void_ratio = 2.14 = 1.3375 \times 64/40`, i.e. the default rescaled to the larger box, while seeds 2–4 record the unused default beside void ratios above it, i.e. they were gated on `-max-void-per-L`. Both are the same size-independent criterion, and all four REV packings sit at 2.1–2.5% of L against the 3.3–3.4% the accepted pilot packings reach.

2.2 Why additive, and why contact is ϵ -independent. Each grain’s own term in (1) crosses 0.5 at exactly $r_k = R_k$ — the sharp grain surface — *for any* ϵ . Since drop-and-roll seats every deposited grain in exact tangency on its supports, neighbours touch precisely where $\phi_1 = \phi_2 = 0.5$. Three consequences:

- contact has an ϵ -independent meaning;
- the radii never depend on ϵ , so the same `grains.dat` is valid at every temperature and resolution;
- no neck calibration step is needed — the sharp geometry is fixed and only the skin thickness changes.

The alternative form, $\tanh$ of the signed distance to the grains’ union (`-ic_grain_union 1`), exists for *overlapping* grains. These packings deliberately have none, so it is not used here.

2.3 Temperature and vapour. Every IC shares one closure: temperature is the linear background field $T = T_0 + \nabla T_0 \cdot (\mathbf{x} - \mathbf{L}/2)$ (the k_{eff} campaign sets no gradient, so $T \equiv T_0$), and vapour density is saturated inside ice and h_0 -scaled in air, blended by the ice fraction:

$$\rho_v = \rho_{vs}(T) [h_0 a + (1 - a)], \quad a = \max(0, 1 - \phi), \quad (2)$$

with $h_0 = 1$ (saturated) for these runs.

2.4 Where ϕ is evaluated. Node coordinates come from the Greville abscissae of the B-spline basis, with the periodic-axis node spacing using N intervals rather than $N - 1$ (a periodic node at $x = L$ coincides with the one at $x = 0$). Getting that denominator wrong stretched every IC by $(N + 1)/N$ and left a double-width gap at the seam — measured as a $2/N$ disc-area excess in `studies/keff_sharp_limit/`.

2.5 ϵ and the mesh. ϵ is not chosen by eye. `preprocess/comp_eps.py` takes the minimum of the three Kaempfer & Plapp (2009) sharp-interface bounds (with safety $s = \frac{1}{2}$) and the geometric bound $0.05 R_{ave}$, with the normal velocity in the kinetic bound derived self-consistently from a feature size $R_{feat} = 2 \mu\text{m}$ so that the bound cannot collapse at cold temperatures. At $T = -20^\circ\text{C}$ this gives $\epsilon = 1 \mu\text{m}$ for these packings, with the heat bound binding. The mesh follows:

$$h = \frac{\epsilon}{\sqrt{2}}, \quad N_x = \left\lceil \frac{\sqrt{2} L_x}{\epsilon} \right\rceil, \quad (3)$$

i.e. 2829^2 for the pilot and 4526^2 for the REV box, i.e. 24M and 61M degrees of freedom at 3 fields per node. The geometry `.opts` also carries `-eps_valid_temp`, and the solver aborts if the run temperature disagrees with it by more than 1°C — ϵ and the mesh are temperature-specific even though the packing is not.

4 What was actually built

Realized values from `metadata.json`, all at $\phi_{target} = 0.325$, $R_{ave} = 50 \mu\text{m}$, `periodic xy`:

set	seed	grains	filler	ϕ	Z	Z_{band}	roll [rad]	rows
pilot, $L = 2.0$ mm	1	309	14	0.3251	1.96	3.40	0.957	334
	2	308	15	0.3251	1.99	3.49	0.982	337
	3	336	21	0.3249	1.99	3.41	1.006	366
	4	310	20	0.3252	1.92	3.36	1.080	336
REV, $L = 3.2$ mm	1	797	27	0.3248	1.98	3.57	1.104	845
	2	805	32	0.3254	2.03	3.53	1.115	844
	3	724	37	0.3249	1.93	3.52	1.264	758
	4	769	42	0.3254	1.95	3.33	0.945	815

`rows` is the line count in `grains.dat`, i.e. distinct grains plus periodic edge images. Two coordination numbers are reported because they answer different questions: Z counts near-exact contacts (≈ 2 , barely rigid), while Z_{band} counts every pair closer than the diffuse band 9.2ϵ (≈ 3.4 , nearly isostatic). **Z_{band} is the number that describes the simulation**, because the solver joins with solid every pair closer than the band. The solid percolates in both axes in all eight packings (largest cluster 75–96% of the solid).

5 Two consequences to keep in view

The diffuse band closes narrow throats at $t = 0$. Because (1) is additive, two grains that nearly touch each contribute ≈ 0.5 at the midpoint of the gap and *sum* to ≈ 1 . At the centre of a gap g the sum is $1 - \tanh(g/4\epsilon)$, so ϕ only falls below 0.01 once

$$g > 4\epsilon \operatorname{artanh}(0.99) \approx 10.6 \epsilon. \quad (4)$$

At $\epsilon = 1 \mu\text{m}$ that is $10.6 \mu\text{m}$, against a median pore throat of $19.4 \mu\text{m}$ (pilot) / $15.6 \mu\text{m}$ (REV) and a 25th percentile of $\approx 1.5 \mu\text{m}$. So roughly a quarter of the throats hold no pore at all at $t = 0$, and a channel narrower than the band conducts like solid however open the drawing looks. This is recorded in the header of every geometry `.opts` and is why the deliverable of the campaign is *comparative* — how much sintering changes k_{eff} — and not absolute k_{eff} in $\text{W m}^{-1} \text{K}^{-1}$. The same effect puts the usable (band-eroded) pore space in hundreds of disconnected pockets at $t = 0$, which bears on vapour transport rather than on conduction.

$t = 0$ is not the baseline; $t = 1$ day is. Equation (1) is an analytic clamped sum, not an equilibrated phase field, and the grains start in exact *point* tangency with no neck. The first hours of every run are therefore the field relaxing rather than the microstructure sintering, and it is precisely at $t = 0$ that a band bridging a point contact does the most damage to k_{eff} : on pilot seed 1, $k_{\text{eff}}(0)$ reads 0.6312 under the arithmetic conductivity law against 0.3695 under the tensor law, i.e. the arithmetic law was +71% high on tangent grains. All reported changes are measured against the $t = 1$ day state.

6 Reproducing it

The generation commands are not committed; the following are reconstructed from `metadata.json` and produce the committed packings (the generator is deterministic in the seed). The flag `-no-periodic-subdir` is what puts the output at the committed path rather than under a `periodic_xy/` level.

```
python3 preprocess/generate_packing_gravity.py \
-Lx 2.0e-3 -porosity 0.325 -mean-r 50e-6 -sigma-ln 0.5 \
-periodic xy -band-per-mean-r 0.184 -seed 1 \
-no-periodic-subdir \
-out inputs/packings/pilot_LR40/pilot_phi0.325_Rave50um_LR40_seed1
```

For the REV set, $-Lx\ 3.2e-3$ and the size-independent void gate $-max-void-per-L\ 0.0335$. The `.opts` matrix on top of a packing comes from `preprocess/generate_study_opts.py` with `-R_feat 2e-6`, and runs go through the run scripts (`scripts/HPC/submit_enceladus.sh <geometry> <experiment> [tag]`), never the bare executable, so the staged inputs and figures exist afterwards.

Seeing the initial condition the solver actually uses. The generator's `preview.png` draws sharp discs, which is the geometry as specified rather than the field the run starts from. To render (1) itself, including a zoom far enough in to make the $9.2\ \mu\text{m}$ band visible at the 2 mm domain scale (it is 0.46% of the frame):

```
venv_enceladus/bin/python studies/keff_sintering/render_pilot_ic.py
```